

Question

The chloride **A** of a first-row transition metal **M** is a red paramagnetic liquid with a tetrahedral molecular geometry and a magnetic moment of $\mu = 1.7B.M$. When **A** is heated in an inert atmosphere, it releases chlorine gas and forms compound **B**; in **B**, chlorine forms a hexagonal close-packed structure, and metal atoms fill one-third of the octahedral voids. Under a high-pressure carbon monoxide atmosphere, in the presence of bis(2-methoxyethyl) ether (which can be abbreviated as **L**), **B** is reduced by metallic Na to obtain a 1:1 salt **C**; the cation of **C** is an octahedral coordination ion of Na, and the anion is an octahedral coordination ion of **M**. **C** reacts with a large amount of dilute phosphoric acid, releasing hydrogen gas and obtaining a green solid **D**; **D** is paramagnetic, sublimable, and easily ignites. **D** undergoes disproportionation in tetrahydrofuran (which can be abbreviated as THF), and the disproportionation product **E** can be recrystallized from dichloromethane; **E** is a 1:2 salt, and its cation is connected to 2 anions through the electrostatic interaction of $M \cdots O$, so that all **M** in its structure are octahedrally coordinated. Based on the above information, select the correct option below.

- A. All other options are incorrect
- B. **A** contains 2 unpaired electrons
- C. **M**'s common oxides do not include M_2O_5
- D. **B** contains 2 chlorine atoms.
- E. **C**'s cation contains three diglyme molecules.
- F. The anion of **C** possesses strong oxidizing properties.
- G. The structure of **D** is octahedral and contains four elements.
- H. The anion in **E** is not the same as the anion in **C**.

- I. In **E**, for the **M** with the higher oxidation state, the ligands attached to it have two different coordination environments

Answer

Correct Answer: **I**

Detailed Explanation

A being a tetrahedral configuration chloride indicates that the chemical formula of **A** is MCl_4 . According to the magnetic moment calculation formula $\mu = \sqrt{n(n+2)}$, the number of unpaired electrons is 1, so option **B** is incorrect.

CHECKPOINT

1 PTS

The number of unpaired electrons is calculated to be 1 based on the magnetic moment

Tetrahedral fields are usually high-spin, and since the number of unpaired electrons is 1, the metal **M** is in the +4 valence state. Therefore, it is inferred that the metal itself has 5 valence layer electrons, so **M** is **V**. **V** has a common oxide V_2O_5 , so option **C** is incorrect.

CHECKPOINT

1 PTS

It is inferred that **M** is **V**

B is obtained from **A** by releasing chlorine gas, so it should also be a chloride of **V**. At the same time, **V** occupies 1/3 of the octahedral voids formed by Cl , so **B** is VCl_3 , and option **D** is incorrect.

CHECKPOINT

1 PTS

It is inferred that **B** is VCl_3

In **C**, the cations and anions are octahedrally coordinated and formed in a high-pressure CO atmosphere. It is speculated that the anion is the CO hexa-coordinated V. The cation is Na coordinated by the ligand **L**. The ligand **L** is a tridentate ligand, so the cation chemical formula is NaL_2^+ , and the chemical formula of **C** is $[\text{NaL}_2][\text{V}(\text{CO})_6]$.

CHECKPOINT

2 PTS

It is inferred that **C** is $[\text{NaL}_2][\text{V}(\text{CO})_6]$

C reacts with acid to release hydrogen gas to generate **D**, indicating that **C** is oxidized. It should be $\text{V}(\text{CO})_6^-$ losing electrons to generate $\text{V}(\text{CO})_6$. In option **G**, it is correct that **D** has an octahedral configuration, but it only contains three elements. **G** is incorrect.

CHECKPOINT

1 PTS

It is inferred that **D** is $\text{V}(\text{CO})_6$

D undergoes a disproportionation reaction to generate **E**. The anion should be $\text{V}(\text{CO})_6^-$. Since the ratio of cations to anions is 2:1, the cation center is V^{2+} . Also, because the cation is connected to the O of two anions, there are four remaining coordination sites, which should be 4 THF. Therefore, **E** is $[\text{V}(\text{THF})_4][\text{V}(\text{CO})_6]_2$. So, option **E** states that the cation only contains two diethylene glycol dimethyl ethers, which is incorrect. In option **H**, **E** and **C** have the same anion, which is incorrect.

In option **F**, $\text{V}(\text{CO})_6^-$ has a negative charge and has strong reducing properties rather than oxidizing properties, so **F** is incorrect.

The structure of **E** is linear, with the cation coordinated by 4 tetrahydrofuran molecules sandwiched between two anionic complexes. One CO from each of the two $\text{V}(\text{CO})_6^-$ participates in the coordination of the central cation. There are two types of O environments, THF and CO, participating in the cation coordination, so option **I** is correct.

CHECKPOINT

2 PTS

It is inferred that **E** is $[\text{V}(\text{THF})_4][\text{V}(\text{CO})_6]_2$

CHECKPOINT

2 PTS

The structure of **E** is linear, with the cation coordinated by 4 tetrahydrofuran molecules sandwiched between two anionic complexes. One CO from each of the two $\text{V}(\text{CO})_6^-$ participates in the coordination of the central cation